Anthony James Padavano

SENIOR SOFTWARE ENGINEER (SYSTEMS & STATE ESTIMATION)

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LINKS

Portfolio · GitHub · agentic-titan · ORGANVM System

PROFILE

Architect of a 113-repository infrastructure spanning 8 GitHub organizations: 82K files, 2,349 tests, 104 CI/CD pipelines, and a formal state machine governing entity lifecycle transitions. Builds high-reliability software in Python and TypeScript with emphasis on state estimation, dependency validation, and data fusion from heterogeneous sources. Automated verification matrices ensure system-wide correctness across interdependent subsystems. Full-Stack Developer (Meta). LGBTQ+.

TECHNICAL SKILLS

Python · State Estimation · Systems Architecture · CI/CD · Docker · Linux · Git · Data Fusion · Performance Analysis · YAML/JSON Schema · pytest · Verification Matrices · GitHub Actions · TypeScript · Node.js · Probabilistic Scoring · CLI Tooling · GCP / Terraform · Redis · WebSocket

SELECTED PROJECTS

ORGANVM Eight-Organ System — 113-Repository State Machine Infrastructure
8 GitHub orgs, 82K files, 2,349 tests, 104 CI/CD pipelines. Formal promotion state machine (LOCAL → CANDIDATE → PUBLIC → GRADUATED) with dependency validation across 43 directed edges. Forward-only transitions and cross-subsystem integrity checks. TypeScript + Python.

Application Pipeline — Multi-Source Scoring & Tracking Engine
Python CLI suite fusing data from heterogeneous sources (APIs, network graphs, outreach logs, market intelligence) into a 9-dimension scoring rubric. YAML state machine tracks entities through 8 lifecycle stages with staleness detection and threshold gating.

agentic-titan — Multi-Agent Orchestration & Tracking System
TypeScript framework coordinating autonomous agents across 18 phases with 1,095+ tests. Real-time task decomposition, parallel execution tracking, and structured state management.

Network Graph — Probabilistic Path-Finding & Scoring
Graph-based scoring engine: hop-decay functions (Granovetter weak-ties), BFS/DFS path-finding, tie-strength tracking. Fuses contact data and outreach signals into org-level proximity scores.

Verification Matrix — System-Wide Correctness Validation
Module-to-test coverage enforcement across 117 modules. IRA evaluation framework (ICC, Cohen's kappa, Fleiss kappa) using multi-model AI panels for quality assessment.

EXPERIENCE

Software Engineer & Systems Architect — 2020 — Present
ORGANVM

- **State Machine Architecture:** Formal entity lifecycle state machine governing 113 repos — forward-only transitions, dependency validation across 43 directed edges, cross-org integrity enforcement.
- **Multi-Source Data Fusion:** Scoring engine fusing signals from heterogeneous sources (APIs, network graphs, market intelligence, outreach logs) into a 9-dimension probabilistic assessment with configurable weighting.
- **Testing & Verification:** 2,349 tests with verification matrix enforcing module-to-test coverage (117 modules). System-wide correctness validation through CI/CD pipelines spanning 8 organizations.
- **Reliability Infrastructure:** 128 GitHub Actions workflows (18 reusable templates), 104 CI/CD pipelines, automated staleness detection, real-time health monitoring across distributed subsystems.

Instructor — Composition & Technical Communication — 2015 — Present
Miami Dade College, Keiser University, Nova Southeastern, FAU, Young Harris, Tarrant County, Broward, Palm Beach State

- 100+ courses across 8 institutions; 2,000+ students. Structured problem decomposition, documentation, and mentorship.

Digital Marketing Manager — 2023 — 2024
Miami Dade College Foundation (nonprofit)

- Rebuilt foundation website; designed measurement frameworks increasing donor engagement by 32%.

Multimedia Specialist — 2011 — 2020
AJP Media Arts — clients incl. Boca Ballet Theatre, Forward Funding, Arts in Boca

- Video, audio, and visual production pipelines for nonprofits and arts organizations; 17.5M+ views, $2M in fundraising.

EDUCATION

M.F.A. Creative Writing, Florida Atlantic University — 2015 — 2018
B.A. English Literature, CUNY College of Staten Island — 2010 — 2014

CERTIFICATIONS

Full-Stack Developer, Meta (2024) · UX Design, Google (2023) · Digital Marketing, Google (2023) · Project Management, Google (2023)